As mentioned, this construction method for k-d trees naturally generalizes to higher dimensions, although it becomes more difficult to visualize the trees and the spaces itself.

For k=3, we can still imagine $\mathbb{R}^3$ divided in parallelepipeds, as shown in figure 9.7, but for 4-D and further, we lack an immediate geometric interpretation; that said, as long as we treat k-D points as tuples, it is possible to follow the same steps we have seen for a 2d-tree, without any change.

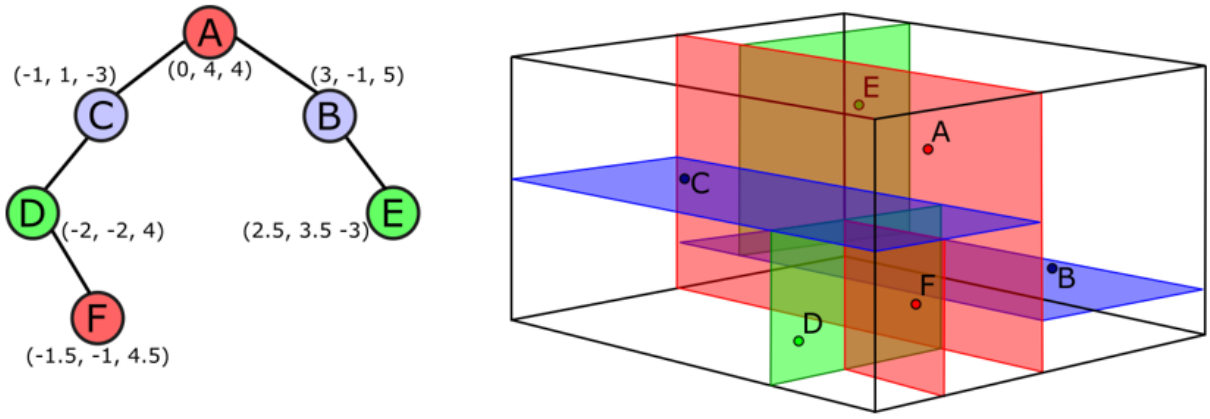


Figure 9.7 An example of a 3-D tree (aka k-d tree with dimension 3). For the sake of clarity, regions are not highlighted, and nodes are filled with the same color of their split planes.

9.2.2 Invariants

We could sum up the definition of k-d trees in a few invariants.

A k-d tree is then defined as a binary search tree, whose elements are k-dimensional points, and that abides by the